Exercises:

- *Show that $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is hermitian and anticommutes with all the γ^μ. Show that in the Dirac-Pauli representation it takes the form (in 2×2 block notation)*

$$\gamma^5 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} .$$

Show that, in the limit $m \to 0$, the solutions of the Dirac equation with definite helicity

$$u(p)_\pm = \sqrt{E_p + m} \begin{pmatrix} \chi_\pm \\ \frac{\vec\sigma\cdot\vec p}{E_p+m}\chi_\pm \end{pmatrix} \quad \text{and} \quad v(p)_\pm = \sqrt{E_p + m} \begin{pmatrix} \frac{\vec\sigma\cdot\vec p}{E_p+m}\chi_\mp \\ \chi_\mp \end{pmatrix}$$

are eigenvectors of γ^5 and show that the eigenvalues (which are called their "chiralities") have the same signs as their helicities for particles, and the opposite signs for antiparticles. Recall that the helicity spin-1/2 wavefuctions satisfy (note that $\hat p$ and not $\vec p$ appears on the left-hand side)

$$\vec\sigma\cdot\hat p\,\chi_\pm = \pm\chi_\pm .$$

We recall that γ^0 is hermitian while $\vec\gamma$ is antihermitian. Thus

$$\begin{aligned}
\gamma^{5\dagger} &= -i(-\gamma^3)(-\gamma^2)(-\gamma^1)(\gamma^0) \\
&= +i\gamma^3\gamma^2\gamma^1\gamma^0 \\
&= -i\gamma^2\gamma^1\gamma^0\gamma^3 \\
&= -i\gamma^1\gamma^0\gamma^2\gamma^3 \\
&= +i\gamma^0\gamma^1\gamma^2\gamma^3 = \gamma^5
\end{aligned}$$

The explicit form is

$$\begin{aligned}
\gamma^5 &= i\begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}\begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_2 \\ -\sigma_2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix} \\
&= i\begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}\begin{pmatrix} -i\sigma_3 & 0 \\ 0 & -i\sigma_3 \end{pmatrix}\begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix} \\
&= \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}\begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} .
\end{aligned}$$

In the limit $m \to 0$, the spinors become

$$u(p)_\pm = \sqrt{E_p}\begin{pmatrix} \chi_\pm \\ \pm\chi_\pm \end{pmatrix} \quad \text{and} \quad v(p)_\pm = \sqrt{E_p}\begin{pmatrix} \mp\chi_\mp \\ \chi_\mp \end{pmatrix} .$$

These are manifestly eigenvectors of γ^5 with chiralities ± 1 for $u(p)_\pm$ and ∓ 1 for $v(p)_\pm$.

Problems:

1. Helicity spin-1/2 wavefunctions. *The defining equation for these wavefunctions is*

$$\vec{\sigma} \cdot \hat{p}\, \chi(\hat{p})_{\pm} = \pm\chi(\hat{p})_{\pm} \,, \tag{1}$$

where now it has been made explicit that these wavefunctions depend on $\hat{p}$. We know that (making conventional choices of overall phases)

$$\chi(\hat{z})_{+} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \chi(\hat{z})_{-} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \,.$$

(a) *Show that*

$$\chi(\hat{p})_{\pm} = R_z(\phi) R_y(\theta) \chi(\hat{z})_{\pm} \,, \tag{2}$$

satisfy the helicity equation (1). Here the rotation matrix about an axis $\hat{n}$ by angle δ is

$$R_n(\delta) = \exp(-i\delta\hat{n} \cdot \vec{\sigma}/2) = \cos(\delta/2)\mathbb{1} - i\sin(\delta/2)\hat{n} \cdot \vec{\sigma} \,.$$

Here θ and ϕ are, respectively, the polar and azimuthal angles of $\hat{p}$ relative to the coordinate axes. You can do this by explicit evaluation or using some of the rotation methodology you learned in graduate QM.

Method 1: using spinology. The key ingredient is the transformation of spin-1/2 operators transform under rotations:

$$\vec{\sigma} \cdot \hat{p} = R_z(\phi) R_y(\theta) \vec{\sigma} \cdot \hat{z} R_y(-\theta) R_z(-\phi) \,.$$

See, for example, Sakurai, *Modern QM*, Eq. (3.2.47). Applying this $\chi(\hat{p})_{\pm}$ as given in Eq. (2), we find

$$\begin{aligned} \vec{\sigma} \cdot \hat{p} \chi(\hat{p}) \pm &= R_z(\phi) R_y(\theta) \vec{\sigma} \cdot \hat{z} R_y(-\theta) R_z(-\phi) R_z(-\phi) R_y(-\theta) \chi(\hat{z})_{\pm} \\ &= R_z(-phi) R_y(\theta) \vec{\sigma} \cdot \hat{z} \chi(\hat{z})_{\pm} \\ &= \pm R_z(\phi) R_y(\theta) \chi(\hat{z})_{\pm} \\ &= \pm\chi(\hat{p})_{\pm} \,. \end{aligned}$$

Method 2: direct evaluation. First we write out the claimed form in detail

$$\chi(\hat{p})_{\pm} = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} \bar{c}_\theta & -\bar{s}_\theta \\ \bar{s}_\theta & \bar{c}_\theta \end{pmatrix} \chi(\hat{z})_{\pm}$$

where $\bar{c}_\theta \equiv \cos(\theta/2)$ and $\bar{s}_\theta \equiv \sin(\theta/2)$. Evaluating the matrix products we find

$$\chi(\hat{p})_{+} = \begin{pmatrix} e^{-i\phi/2}\bar{c}_\theta \\ e^{i\phi/2}\bar{s}_\theta \end{pmatrix} = e^{i\phi/2}\bar{c}_\theta \begin{pmatrix} e^{-i\phi} \\ \bar{t}_\theta \end{pmatrix} \tag{3}$$

$$\chi(\hat{p})_{-} = \begin{pmatrix} -e^{-i\phi/2}\bar{s}_\theta \\ e^{i\phi/2}\bar{c}_\theta \end{pmatrix} = -e^{i\phi/2}\bar{s}_\theta \begin{pmatrix} e^{-i\phi} \\ -1/\bar{t}_\theta \end{pmatrix} \,, \tag{4}$$

where $\bar{t}_\theta = \tan(\theta/2)$. Now we need to express $\bar{t}_\theta$ and $e^{-i\phi}$ in terms of the components of $\hat{p}$. First we use

$$c_\theta = \hat{p}_3 \quad \Rightarrow \bar{t}_\theta^2 = \frac{1-c_\theta}{1+c_\theta} = \frac{1-\hat{p}_3}{1+\hat{p}_3} = \frac{p_{12}^2}{(1+\hat{p}_3)^2} = \frac{(1-\hat{p}_3)^2}{p_{12}^2}\,,$$

where $p_{12}^2 = \hat{p}_1^2 + \hat{p}_2^2 = 1 - \hat{p}_3^2$. Since $0 \leq \theta/2 \leq \pi/2$ we know that $\bar{t}_\theta \geq 0$ so we should take the positive square root

$$\bar{t}_\theta = \frac{p_{12}}{1+\hat{p}_3} = \frac{1-\hat{p}_3}{p_{12}}\,.$$

From the definition of ϕ we have

$$e^{-i\phi} = \cos\phi - i\sin\phi = \frac{\hat{p}_1}{p_{12}} - i\frac{\hat{p}_2}{p_{12}} = \frac{\hat{p}_1 - i\hat{p}_2}{p_{12}}\,.$$

Combining all the above we find that

$$\chi(\hat{p})_+ \propto \begin{pmatrix} \hat{p}_1 - i\hat{p}_2 \\ 1-\hat{p}_3 \end{pmatrix} \quad \text{and} \quad \chi(\hat{p})_- \propto \begin{pmatrix} \hat{p}_1 - i\hat{p}_2 \\ -(1+\hat{p}_3) \end{pmatrix}\,.$$

Finally we write out $\vec{\sigma}\cdot\hat{p}$ explicitly

$$\vec{\sigma}\cdot\hat{p} = \begin{pmatrix} \hat{p}_3 & \hat{p}_1 - i\hat{p}_2 \\ \hat{p}_1 + i\hat{p}_2 & -\hat{p}_3 \end{pmatrix}\,,$$

from which it is easy to check that the eigenvalues of $\chi(\hat{p})_\pm$ are ± 1.

(b) *Evaluate*

$$\chi(\hat{p})_\pm^\dagger \chi(\hat{z})_{\pm'}$$

for each of the four possible choices of helicities. Using the results (3) and (4) from above, plus the forms of $\chi(\hat{z})_\pm$, we find

$$\begin{aligned}
\chi(\hat{p})_+^\dagger \chi(\hat{z})_+ &= e^{i\phi/2}\bar{c}_\theta\,,\\
\chi(\hat{p})_-^\dagger \chi(\hat{z})_+ &= -e^{i\phi/2}\bar{s}_\theta\,,\\
\chi(\hat{p})_+^\dagger \chi(\hat{z})_- &= e^{-i\phi/2}\bar{s}_\theta\,,\\
\chi(\hat{p})_-^\dagger \chi(\hat{z})_- &= e^{-i\phi/2}\bar{c}_\theta\,.
\end{aligned}$$

These results were used to calculate the Yukawa scattering matrix elements in class.

(c) *Show that the spin-averaged outer products of spinor solutions satisfy*

$$\begin{aligned}
\mathcal{U}_{\alpha\beta} &\equiv u(p)_{+,\alpha}\bar{u}(p)_{+,\beta} + u(p)_{-,\alpha}\bar{u}(p)_{-,\beta} = \left[\not{p} + m\right]_{\alpha\beta}\\
\mathcal{V}_{\alpha\beta} &\equiv v(p)_{+,\alpha}\bar{v}(p)_{+,\beta} + v(p)_{-,\alpha}\bar{v}(p)_{-,\beta} = \left[\not{p} - m\right]_{\alpha\beta}\,.
\end{aligned}$$

Here α and β are spinor indices that run from $1-4$. These results are used in evaluating spin-averaged cross sections. Recall that

$$u(p)_\pm = \sqrt{E_p+m}\begin{pmatrix} \chi(\hat{p})_\pm \\ \pm\frac{|\vec{p}|}{E_p+m}\chi(\hat{p})_\pm \end{pmatrix} \quad \text{and} \quad v(p)_\pm = \sqrt{E_p+m}\begin{pmatrix} \mp\frac{|\vec{p}|}{E_p+m}\chi(\hat{p})_\mp \\ \chi(\hat{p})_\mp \end{pmatrix}.$$

Evaluating the outer product for particle spinors, and recalling that $\bar{u} = u^\dagger\gamma^0$, we find

$$\mathcal{U} = \begin{pmatrix} (E_p+m)\mathcal{M}_+ & -|\vec{p}|\mathcal{M}_- \\ |\vec{p}|\mathcal{M}_- & -\frac{\vec{p}^2}{(E_p+m)}\mathcal{M}_+ \end{pmatrix}, \tag{5}$$

where the spin-wavefunction outer products are

$$\mathcal{M}_\pm = \chi(\hat{p})_+\chi(\hat{p})_+^\dagger \pm \chi(\hat{p})_-\chi(\hat{p})_-^\dagger .$$

We evaluate $\mathcal{M}_+$ using Eq. (2) together with

$$\chi(\hat{p})_\pm^\dagger = \chi(\hat{z})_\pm^\dagger R_y(-\theta)R_z(-\phi)$$

and

$$\chi(\hat{z})_+\chi(\hat{z})_+^\dagger + \chi(\hat{z})_-\chi(\hat{z})_-^\dagger = \mathbb{1},$$

finding

$$\mathcal{M}_+ = R_z(-\phi)R_y(-\theta)\mathbb{1}R_y(\theta)R_z(\phi) = \mathbb{1}.$$

To evaluate $\mathcal{M}_-$ we use

$$\vec{\sigma}\cdot\hat{p}\mathcal{M}_- = \mathcal{M}_+ = \mathbb{1} \quad \Rightarrow \quad \mathcal{M}_- = \vec{\sigma}\cdot\hat{p},$$

since $(\vec{\sigma}\cdot\hat{p})^2 = \mathbb{1}$. Inserting the results for $\mathcal{M}_\pm$ into Eq. (5) we find

$$\mathcal{U} = \begin{pmatrix} E_p+m & -\vec{\sigma}\cdot\vec{p} \\ \vec{\sigma}\cdot\vec{p} & -E_p+m \end{pmatrix} = \not{p} + m.$$

Now we repeat the exercise for the antiparticle spinors. Using their explicit form we find

$$\begin{aligned} \mathcal{V} &= \begin{pmatrix} (E_p-m)\mathcal{M}_+ & -|\vec{p}|\mathcal{M}_- \\ |\vec{p}|\mathcal{M}_- & -(E_p+m)\mathcal{M}_+ \end{pmatrix} \\ &= \begin{pmatrix} E_p-m & -\vec{\sigma}\cdot\vec{p} \\ \vec{\sigma}\cdot\vec{p} & -E_p-m \end{pmatrix} \\ &= \not{p} - m. \end{aligned}$$

2. Dirac traceology. *In class we showed, using only the defining anticommutation properties of gamma matrices and the properties of γ^5, that the trace of any odd number of gamma matrices vanishes and that*

$$\mathrm{tr}(\not{p}\not{q}) = 4p\cdot q.$$

Here we extend these results.

(a) *Show that*

$$\mathrm{tr}\left[\left(\gamma^0\right)^{n_0}\left(\gamma^1\right)^{n_1}\left(\gamma^2\right)^{n_2}\left(\gamma^3\right)^{n_3}\right]$$

vanishes unless the positive integers n_0, n_1, n_2 and n_3 are all even. Note that this is a stronger than the result quoted above that the trace of an odd number of gamma matrices must vanish. It must be that the power of **each** *of the gamma matrices separately must be even. Furthermore, since the different gamma matrices anticommute, this statement holds independent of the order of various gamma matrices.*

Since $(\gamma^\mu)^2 = \pm\mathbb{1}$, we need only consider the cases $n_\mu = 0$ or 1. Since the trace of any odd number of gamma matrices vanishes, this leaves only the following cases to consider

$$\mathrm{tr}(\gamma^\mu\gamma^\nu) \;\; [\text{with } \mu \neq \nu] \quad \text{and} \quad \mathrm{tr}(\gamma^0\gamma^1\gamma^2\gamma^3) \propto \mathrm{tr}(\gamma^5)\,.$$

The former vanishes using the cyclicity of the trace and the defining property of gamma matrices

$$\mathrm{tr}(\gamma^\mu\gamma^\nu) = \mathrm{tr}(\gamma^\nu\gamma^\mu) = \frac{1}{2}\mathrm{tr}\left([\gamma^\mu,\gamma^\nu]_+\right) = 4g^{\mu\nu} = 0 \text{ for } \mu \neq \nu\,.$$

The latter vanishes using the fact that γ^5 anticommutes with all the γ^μ:

$$\mathrm{tr}(\gamma^5) = \mathrm{tr}(\gamma^0\gamma^0\gamma^5) = -\mathrm{tr}(\gamma^0\gamma^5\gamma^0) = -\mathrm{tr}(\gamma^5) = 0\,.$$

(b) *Evaluate (providing explanation for your work)*

$$\mathrm{tr}(\not{p}_1\not{p}_2\not{p}_3\not{p}_4) \;\; \text{and} \;\; \mathrm{tr}(\gamma^5\not{p}_1\not{p}_2\not{p}_3\not{p}_4)\,.$$

To evaluate the first trace we move the $\not{p}_1$ through the trace until we return to the initial position. We need

$$\not{p}_1\not{p}_2 = -\not{p}_2\not{p}_1 + 2p_1\cdot p_2\mathbb{1}\,,$$

which follows from the defining anticommutation relation of gamma matrices. Thus we have

$$\begin{aligned}
\mathrm{tr}(\not{p}_1\not{p}_2\not{p}_3\not{p}_4) &= -\mathrm{tr}(\not{p}_2\not{p}_1\not{p}_3\not{p}_4) + 8p_1\cdot p_2 p_3\cdot p_4 \\
&= \mathrm{tr}(\not{p}_2\not{p}_3\not{p}_1\not{p}_4) + 8p_1\cdot p_2 p_3\cdot p_4 - 8p_1\cdot p_3 p_2\cdot p_4 \\
&= -\mathrm{tr}(\not{p}_1\not{p}_2\not{p}_3\not{p}_4) + 8p_1\cdot p_2 p_3\cdot p_4 - 8p_1\cdot p_3 p_2\cdot p_4 + 8p_1\cdot p_4 p_2\cdot p_3\,.
\end{aligned}$$

Here at each stage we have used $\mathrm{tr}(\not{p}_1\not{p}_2) = 4p_1\cdot p_2$. Now we have the same trace on both sides, but with opposite signs, so we can recombine these terms and divede by two leading to

$$\mathrm{tr}(\not{p}_1\not{p}_2\not{p}_3\not{p}_4) = 4p_1\cdot p_2 p_3\cdot p_4 - 4p_1\cdot p_3 p_2\cdot p_4 + 4p_1\cdot p_4 p_2\cdot p_3\,.$$

Now we turn to

$$\mathrm{tr}(\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = \mathrm{tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) p_{1,\mu} p_{2,\nu} p_{3,\rho} p_{4,\sigma} \,.$$

Since $\gamma^5 = ii\gamma^0\gamma^1\gamma^2\gamma^3$ we know from the result of the first part of this problem that the trace is nonzero only when the indices μ, ν, ρ and σ are all different. If $\{\mu, \nu, \rho, \sigma\} = \{0, 1, 2, 3\}$ then the trace is $-4i$, since $\mathrm{tr}(\gamma^{5\,2}) = 4$. If we permute the indices, then the anticommutation property of the gamma matrices will lead to a sign corresponding to the signature of the permutation, e.g. $\{1, 0, 2, 3\}$ has signature -1. This totally antisymmetric property is represented by the four-index ϵ tensor. Thus we find

$$\mathrm{tr}(\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = -4i\epsilon^{\mu\nu\rho\sigma} p_{1,\mu} p_{2,\nu} p_{3,\rho} p_{4,\sigma} \,,$$

where we are using the convention

$$\epsilon^{0123} = +1 \,.$$

3. *In lecture, we discussed leading-order $e^-\mu^- \to e^-\mu^-$ scattering in QED, finding the result*

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{e^4}{t^2} \left|\bar{u}_3 \gamma^\mu u_1 \bar{u}_4 \gamma_\mu u_2\right|^2 \,.$$

In this problem we evaluate the spinor matrix elements explicitly using the helicity basis. We work in the CM frame.

   (a) *Using the form of the helicity-basis spinor solutions given above, evaluate the spinor currents*

$$\bar{u}(p_3)_+ \gamma^\mu u(p_1)_+ \,, \quad \bar{u}(p_3)_+ \gamma^\mu u(p_1)_- \,, \quad \bar{u}(p_3)_- \gamma^\mu u(p_1)_+ \,, \quad \text{and} \quad \bar{u}(p_3)_- \gamma^\mu u(p_1)_- \,.$$

   You should take $\hat{p}_1 = (0, 0, 1)$ and $\hat{p}_3 = (s_\theta c_\phi, s_\theta s_\phi, c_\theta)$, where $c_\phi = \cos\phi$, etc..

   For definiteness, let's begin with $\bar{u}(p_3)_+ \gamma^\mu u(p_1)_+$, which we'll abbreviate to $\bar{u}_{3+} \gamma^\mu u_{1+}$. Using the form for the spinors given above we find (using $E_3 = E_1 = E_e$ and $|\vec{p}_1| = |\vec{p}_3| = p^*$)

$$\begin{aligned} \bar{u}_{3+} \gamma^0 u_{1+} &= u_{3+}^\dagger u_{1+} = 2E_e \chi(\hat{p}_3)^\dagger \chi(\hat{p}_1) \\ \bar{u}_{3+} \vec{\gamma}\, u_{1+} &= u_{3+}^\dagger \gamma^0 \vec{\gamma}\, u_{1+} = 2p^* \chi(\hat{p}_3)^\dagger \vec{\sigma} \chi(\hat{p}_1) \,. \end{aligned}$$

   Using Eqs. (3) and (4) we then obtain

$$\bar{u}_{3+} \gamma^\mu u_{1+} = 2\left(E_e e^{i\phi/2} \bar{c}_\theta \,, p^* e^{-i\phi/2} \bar{s}_\theta \,, ip^* e^{-i\phi/2} \bar{s}_\theta \,, p^* e^{i\phi/2} \bar{c}_\theta\right) \,.$$

   [Aside: One way to check this result is to see whether current conservation holds, i.e. whether $(p_3 - p_1)_\mu \bar{u}_3 \gamma^\mu u_1 = 0$. Well,

$$\begin{aligned} \tfrac{1}{2} p_{3,\mu} \left(\bar{u}_{3+} \gamma^\mu u_{1+}\right) &= E_e^2 e^{i\phi/2} \bar{c}_\theta - p^{*2} e^{-i\phi/2} \bar{s}_\theta s_\theta (c_\phi + is_\phi) - p^{*2} e^{i\phi/2} c_\theta \bar{c}_\theta \\ &= m_e^2 e^{i\phi/2} \bar{c}_\theta + p^{*2} e^{i\phi/2} \left[\bar{c}_\theta (1 - c_\theta) - 2\bar{s}_\theta^2 \bar{c}_\theta\right] = m_e^2 e^{i\phi/2} \bar{c}_\theta \,, \end{aligned}$$

which agrees with

$$\tfrac{1}{2}p_{1,\mu}\,(\bar{u}_{3+}\gamma^\mu u_{1+}) = E_e^2 e^{i\phi/2}\bar{c}_\theta - p^{*2}e^{i\phi/2}\bar{c}_\theta = m_e^2 e^{i\phi/2}\bar{c}_\theta\,.]$$

By similar manipulations, one finds

$$\begin{aligned}
\bar{u}_{3+}\gamma^\mu u_{1-} &= 2\left(m_e e^{-i\phi/2}\bar{s}_\theta\,,0\,,0\,,0\right)\,,\\
\bar{u}_{3-}\gamma^\mu u_{1+} &= 2\left(-m_e e^{i\phi/2}\bar{s}_\theta\,,0\,,0\,,0\right)\,,\\
\bar{u}_{3-}\gamma^\mu u_{1-} &= 2\left(E_e e^{-i\phi/2}\bar{c}_\theta\,,p^* e^{i\phi/2}\bar{s}_\theta\,,-ip^* e^{i\phi/2}\bar{s}_\theta\,,p^* e^{-i\phi/2}\bar{c}_\theta\right)\,.
\end{aligned}$$

(b) *Using a parity transformation, or otherwise, determine the corresponding four spinor currents of the form $\bar{u}_4\gamma^\mu u_2$.*

Parity flips 3-momenta and thus leads to $\vec{p}_3 \leftrightarrow \vec{p}_4$ and $\vec{p}_1 \leftrightarrow \vec{p}_2$ in the CM frame. Spins are unaffected by parity, so helicities are flipped. Recalling that parity acts on spinors with γ^0, it follows that

$$u_{4\pm} = \gamma^0 u_{3\mp} \quad\text{and}\quad u_{2\pm} = \gamma^0 u_{1\mp}\,.$$

Using

$$\gamma^0\gamma^\mu\gamma^0 = \gamma_\mu\,,$$

(so that the spatial components are flipped in sign) we then find that

$$\begin{aligned}
\bar{u}_{4-}\gamma^\mu u_{2-} &= 2\left(E_\mu e^{i\phi/2}\bar{c}_\theta\,,-p^* e^{-i\phi/2}\bar{s}_\theta\,,-ip^* e^{-i\phi/2}\bar{s}_\theta\,,-p^* e^{i\phi/2}\bar{c}_\theta\right)\,,\\
\bar{u}_{4-}\gamma^\mu u_{2+} &= 2\left(m_\mu e^{-i\phi/2}\bar{s}_\theta\,,0\,,0\,,0\right)\,,\\
\bar{u}_{4+}\gamma^\mu u_{2-} &= 2\left(-m_\mu e^{i\phi/2}\bar{s}_\theta\,,0\,,0\,,0\right)\,,\\
\bar{u}_{4+}\gamma^\mu u_{2+} &= 2\left(E_\mu e^{-i\phi/2}\bar{c}_\theta\,,-p^* e^{i\phi/2}\bar{s}_\theta\,,+ip^* e^{i\phi/2}\bar{s}_\theta\,,-p^* e^{-i\phi/2}\bar{c}_\theta\right)\,.
\end{aligned}$$

Here we have also replaced E_e with E_μ and m_e with m_μ.

(c) *Taking the limit $m_e, m_\mu \to 0$, determine the quantity*

$$|\bar{u}_3\gamma^\mu u_1\bar{u}_4\gamma_\mu u_2|^2$$

for all helicity choices for which it does not vanish. Can you understand these results in terms of conservation of angular momentum?

In the massless limit (where $E_e = E_\mu = E$), helicity is conserved at each vertex (as discussed in lectures). So there are only four nonvanishing spinor amplitudes:

$$\begin{aligned}
\bar{u}_{3+}\gamma^\mu u_{1+}\,\bar{u}_{4+}\gamma_\mu u_{2+} &= 4E^2(\bar{c}_\theta^2 + \bar{s}_\theta^2 + \bar{s}_\theta^2 + \bar{c}_\theta^2) = 8E^2 = \bar{u}_{3-}\gamma^\mu u_{1-}\,\bar{u}_{4-}\gamma_\mu u_{2-}\\
\bar{u}_{3+}\gamma^\mu u_{1+}\,\bar{u}_{4-}\gamma_\mu u_{2-} &= 8E^2 e^{i\phi}\bar{c}_\theta^2 = \bar{u}_{3-}\gamma^\mu u_{1-}\,\bar{u}_{4+}\gamma_\mu u_{2+}\,.
\end{aligned}$$

Squaring these gives

$$\begin{aligned}
\left|\bar{u}_{3+}\gamma^\mu u_{1+}\,\bar{u}_{4+}\gamma_\mu u_{2+}\right|^2 &= 64E^4 = \left|\bar{u}_{3-}\gamma^\mu u_{1-}\,\bar{u}_{4-}\gamma_\mu u_{2-}\right|^2\\
\left|\bar{u}_{3+}\gamma^\mu u_{1+}\,\bar{u}_{4-}\gamma_\mu u_{2-}\right|^2 &= 16E^4(1+c_\theta)^2 = \left|\bar{u}_{3-}\gamma^\mu u_{1-}\,\bar{u}_{4+}\gamma_\mu u_{2+}\right|^2\,.
\end{aligned}$$

If the two incoming helicities are equal, then the projection of angular momentum along $\vec{p}_1$ vanishes, as does the projection of the final angular momentum along $\vec{p}_3$. Thus neither forward nor backward scattering is favored, and we might expect a uniform (θ-independent) angular distribution.

If the incoming helicities are opposite, then the projection along $\vec{p}_1$ is either ± 1. Since helicity is conserved at the vertices, the same projection holds along $\vec{p}_3$ in the final state. Thus forward scattering is favored, consistent with the $1 + \cos\theta$ angular distribution.

(d) *Using the results obtained above, determine the differential cross section after summing over final helicities and averaging over initial helicities. The result should agree with that obtained in class.*

The result obtained in class for the spin sum/average was (lecture notes page 6.13)

$$\tfrac{1}{4}\sum_{\text{helicities}} |\bar{u}_3\gamma^\mu u_1 \bar{u}_4 \gamma_\mu u_2|^2 = 2(s^2+u^2) \qquad (m_e = m_\mu = 0)\,. \tag{6}$$

The helicity-basis calculation has four nonzero contributions, leading to

$$\begin{aligned}\tfrac{1}{4}\sum_{\text{helicities}} |\bar{u}_3\gamma^\mu u_1 \bar{u}_4 \gamma_\mu u_2|^2 &= \tfrac{1}{4}\left[64E^4 + 64E^4 + 16E^4(1+c_\theta)^2 + 16E^4(1+c_\theta)^2\right] \\ &= 32E^4 + 8E^4(1+c_\theta)^2 \\ &= 2s^2 + 2u^2\,,\end{aligned}$$

where we have used the results (for the massless limit)

$$s = 4E^2 \quad \text{and} \quad u = 2p_1\cdot p_4 = 2E^2(1+\cos\theta)\,.$$

Thus we obtain

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s}\frac{e^4}{t^2} 2(s^2+u^2)\,,$$

in agreement with the lecture notes.